

Software Requirements Specification for Software Engineering: subtitle describing software

Team 11, technically functional

Matthew Huynh

Cieran Diebolt

Vaisnavi Shanthamoorthy

Maham Siddiqui

Eman Ashraf

October 10, 2025

Contents

1	Purpose of the Project	vi
1.1	User Business	vi
1.2	Goals of the Project	vi
2	Naming Conventions and Terminology	vi
3	Stakeholders	vi
3.1	Client	vi
3.2	Customer	vii
3.3	Other Stakeholders	vii
3.4	Hands-On Users of the Project	vii
3.5	Personas	vii
3.6	Priorities Assigned to Users	vii
3.7	User Participation	vii
3.8	Maintenance Users and Service Technicians	viii
4	Mandated Constraints	viii
4.1	Solution Constraints	viii
4.2	Implementation Environment of the Current System	viii
4.3	Partner or Collaborative Applications	viii
4.4	Off-the-Shelf Software	viii
4.5	Anticipated Workplace Environment	viii
4.6	Schedule Constraints	viii
4.7	Budget Constraints	viii
4.8	Enterprise Constraints	viii
5	Naming Conventions and Terminology	ix
5.1	Glossary of All Terms, Including Acronyms, Used by Stake- holders involved in the Project	ix
6	Relevant Facts And Assumptions	ix
6.1	Relevant Facts	ix
6.2	Business Rules	ix
6.3	Assumptions	ix

7	The Scope of the Work	ix
7.1	The Current Situation	ix
7.2	The Context of the Work	ix
7.3	Work Partitioning	ix
7.4	Specifying a Business Use Case (BUC)	ix
8	Business Data Model and Data Dictionary	x
8.1	Business Data Model	x
8.2	Data Dictionary	x
9	The Scope of the Product	x
9.1	Product Boundary	x
9.2	Product Use Case Table	x
9.3	Individual Product Use Cases (PUC's)	x
10	Functional Requirements	x
10.1	Functional Requirements	x
11	Look and Feel Requirements	x
11.1	Appearance Requirements	x
11.2	Style Requirements	xi
12	Usability and Humanity Requirements	xi
12.1	Ease of Use Requirements	xi
12.2	Personalization and Internationalization Requirements	xi
12.3	Learning Requirements	xi
12.4	Understandability and Politeness Requirements	xi
12.5	Accessibility Requirements	xi
13	Performance Requirements	xi
13.1	Speed and Latency Requirements	xi
13.2	Safety-Critical Requirements	xi
13.3	Precision or Accuracy Requirements	xii
13.4	Robustness or Fault-Tolerance Requirements	xii
13.5	Capacity Requirements	xii
13.6	Scalability or Extensibility Requirements	xii
13.7	Longevity Requirements	xii

14 Operational and Environmental Requirements	xii
14.1 Expected Physical Environment	xii
14.2 Wider Environment Requirements	xii
14.3 Requirements for Interfacing with Adjacent Systems	xii
14.4 Productization Requirements	xiii
14.5 Release Requirements	xiii
15 Maintainability and Support Requirements	xiii
15.1 Maintenance Requirements	xiii
15.2 Supportability Requirements	xiii
15.3 Adaptability Requirements	xiii
16 Security Requirements	xiii
16.1 Access Requirements	xiii
16.2 Integrity Requirements	xiii
16.3 Privacy Requirements	xiii
16.4 Audit Requirements	xiv
16.5 Immunity Requirements	xiv
17 Cultural Requirements	xiv
17.1 Cultural Requirements	xiv
18 Compliance Requirements	xiv
18.1 Legal Requirements	xiv
18.2 Standards Compliance Requirements	xiv
19 Open Issues	xiv
20 Off-the-Shelf Solutions	xiv
20.1 Ready-Made Products	xiv
20.2 Reusable Components	xiv
20.3 Products That Can Be Copied	xv
21 New Problems	xv
21.1 Effects on the Current Environment	xv
21.2 Effects on the Installed Systems	xv
21.3 Potential User Problems	xv
21.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product	xv

21.5 Follow-Up Problems	xv
22 Tasks	xv
22.1 Project Planning	xv
22.2 Planning of the Development Phases	xv
23 Migration to the New Product	xvi
23.1 Requirements for Migration to the New Product	xvi
23.2 Data That Has to be Modified or Translated for the New System	xvi
24 Costs	xvi
25 User Documentation and Training	xvi
25.1 User Documentation Requirements	xvi
25.2 Training Requirements	xvi
26 Waiting Room	xvi
27 Ideas for Solution	xvi

Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

Introduction

The software requirements specification (SRS) document describes the functional and non-functional requirements, scope and functionality of the application. It formally outlines key measures relevant to all the different stages of the software development lifecycle. This document is constructed using the Volere Software Requirements Specification template

1 Purpose of the Project

1.1 User Business

Insert your content here.

1.2 Goals of the Project

Insert your content here.

2 Naming Conventions and Terminology

Insert your content here.

3 Stakeholders

The stakeholders for this project are parties with a vested interest in the end product. They will be consulted on their requirements and opinions throughout the development process. Their impact will be foundational for correctness of the product. This section outlines the various types and levels of stakeholders, the personas, user participation (towards application development) and any users that will be responsible for the product's maintenance.

3.1 Client

There are two categories of clients that are the target audience for this product. The first category of end users is the patients. The application is being designed for patients who are undergoing physiotherapy treatment and have been given a home exercise program to aid in their rehabilitation. For the

purpose of this system, the exercise and the joint selected was recommended by one of our stakeholders. The knee and a straight leg raise were chosen.

The second category of the clients are the physiotherapists who are treating the users on the application. It will integrate the physiotherapist's patient list and their corresponding performance statistics. The application allows physiotherapists to gain insight into how their patients' are progressing. Furthermore, it will enable the assessment of the suitability of the prescribed exercise, as well as gauging whether any additional modifications are required (number of repetitions, degree of complexity etc.).

3.2 Customer

The application is being developed for a Canadian market with its governing standards and regulations in mind. Although this version of the application does not have any customers, it can prospectively be purchased by any healthcare provider and be incorporated into their patient care structure.

3.3 Other Stakeholders

The primary and secondary stakeholders are described above (clients and customers).

Other stakeholders may include regulatory authorities or other healthcare specialists, such as physiatrists or massage therapists. The regulatory authorities will be allowed access to conduct audits and ensure that any disclosed patient information is being handled in an ethical manner. In addition, they will verify the accuracy and quality of care being delivered. Other healthcare providers may benefit from any insights or information that is being provided, which may be used to tailor their own care of the patient. Lastly, support workers may use this application to assist their clients with their home exercises.

3.4 Hands-On Users of the Project

Insert your content here.

3.5 Personas

Insert your content here.

3.6 Priorities Assigned to Users

Insert your content here.

3.7 User Participation

Insert your content here.

3.8 Maintenance Users and Service Technicians

Insert your content here.

4 Mandated Constraints

4.1 Solution Constraints

Insert your content here.

4.2 Implementation Environment of the Current System

Insert your content here.

4.3 Partner or Collaborative Applications

Insert your content here.

4.4 Off-the-Shelf Software

Insert your content here.

4.5 Anticipated Workplace Environment

Insert your content here.

4.6 Schedule Constraints

Insert your content here.

4.7 Budget Constraints

Insert your content here.

4.8 Enterprise Constraints

Insert your content here.

5 Naming Conventions and Terminology

5.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

6 Relevant Facts And Assumptions

6.1 Relevant Facts

Insert your content here.

6.2 Business Rules

Insert your content here.

6.3 Assumptions

Insert your content here.

7 The Scope of the Work

7.1 The Current Situation

Insert your content here.

7.2 The Context of the Work

Insert your content here.

7.3 Work Partitioning

Insert your content here.

7.4 Specifying a Business Use Case (BUC)

Insert your content here.

8 Business Data Model and Data Dictionary

8.1 Business Data Model

Insert your content here.

8.2 Data Dictionary

Insert your content here.

9 The Scope of the Product

9.1 Product Boundary

Insert your content here.

9.2 Product Use Case Table

Insert your content here.

9.3 Individual Product Use Cases (PUC's)

Insert your content here.

10 Functional Requirements

10.1 Functional Requirements

Insert your content here.

11 Look and Feel Requirements

11.1 Appearance Requirements

Insert your content here.

11.2 Style Requirements

Insert your content here.

12 Usability and Humanity Requirements

12.1 Ease of Use Requirements

Insert your content here.

12.2 Personalization and Internationalization Requirements

Insert your content here.

12.3 Learning Requirements

Insert your content here.

12.4 Understandability and Politeness Requirements

Insert your content here.

12.5 Accessibility Requirements

Insert your content here.

13 Performance Requirements

13.1 Speed and Latency Requirements

Insert your content here.

13.2 Safety-Critical Requirements

Insert your content here.

13.3 Precision or Accuracy Requirements

Insert your content here.

13.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

13.5 Capacity Requirements

Insert your content here.

13.6 Scalability or Extensibility Requirements

Insert your content here.

13.7 Longevity Requirements

Insert your content here.

14 Operational and Environmental Requirements

14.1 Expected Physical Environment

Insert your content here.

14.2 Wider Environment Requirements

Insert your content here.

14.3 Requirements for Interfacing with Adjacent Systems

Insert your content here.

14.4 Productization Requirements

Insert your content here.

14.5 Release Requirements

Insert your content here.

15 Maintainability and Support Requirements

15.1 Maintenance Requirements

Insert your content here.

15.2 Supportability Requirements

Insert your content here.

15.3 Adaptability Requirements

Insert your content here.

16 Security Requirements

16.1 Access Requirements

Insert your content here.

16.2 Integrity Requirements

Insert your content here.

16.3 Privacy Requirements

Insert your content here.

16.4 Audit Requirements

Insert your content here.

16.5 Immunity Requirements

Insert your content here.

17 Cultural Requirements

17.1 Cultural Requirements

Insert your content here.

18 Compliance Requirements

18.1 Legal Requirements

Insert your content here.

18.2 Standards Compliance Requirements

Insert your content here.

19 Open Issues

Insert your content here.

20 Off-the-Shelf Solutions

20.1 Ready-Made Products

Insert your content here.

20.2 Reusable Components

Insert your content here.

20.3 Products That Can Be Copied

Insert your content here.

21 New Problems

21.1 Effects on the Current Environment

Insert your content here.

21.2 Effects on the Installed Systems

Insert your content here.

21.3 Potential User Problems

Insert your content here.

21.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Insert your content here.

21.5 Follow-Up Problems

Insert your content here.

22 Tasks

22.1 Project Planning

Insert your content here.

22.2 Planning of the Development Phases

Insert your content here.

23 Migration to the New Product

23.1 Requirements for Migration to the New Product

Insert your content here.

23.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

24 Costs

Insert your content here.

25 User Documentation and Training

25.1 User Documentation Requirements

Insert your content here.

25.2 Training Requirements

Insert your content here.

26 Waiting Room

Insert your content here.

27 Ideas for Solution

Insert your content here.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?